

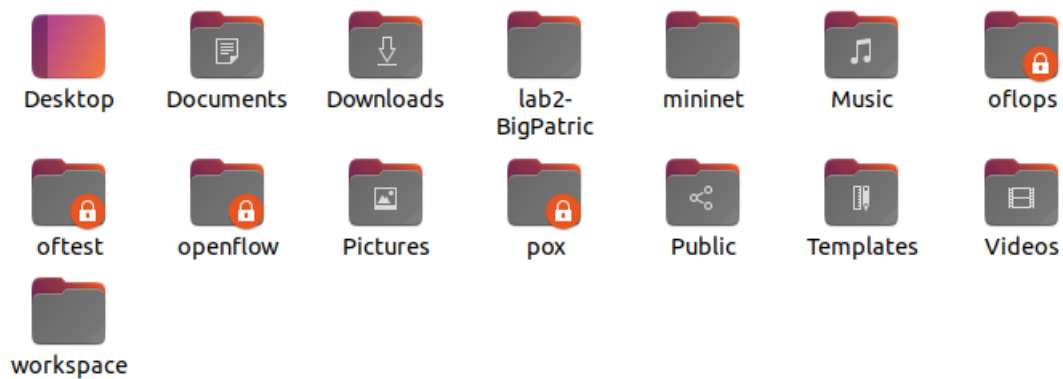
Steps

1.Environment set up :

Following the guidance in the slide and HackMD, the installation of NS3 , this time I use the vscode in my local to finish the coding part(using git push & pull) and execute it in Virtual Box.

2.The new Topo

(1) File part



This is the HOME in virtual box, the *lab2-BigPatric* is the one I clone and get the update of the file and then into the workspace directory, copy the *112550018.cc* file in to the *scratch* directory.

(2) The code

```
NodeContainer nodes;  
nodes.Create(3); // 創建3個節點
```

3 nodes in total

```
PointToPointHelper pointToPoint1, pointToPoint2;  
pointToPoint1.SetDeviceAttribute("DataRate", StringValue("2Mbps"));  
pointToPoint1.SetChannelAttribute("Delay", StringValue("2ms"));  
pointToPoint2.SetDeviceAttribute("DataRate", StringValue("3Mbps"));  
pointToPoint2.SetChannelAttribute("Delay", StringValue("2ms"));
```

In this topo, we will generate two flows, so I create two *pointToPoint*,

with different DataRate

```
NetDeviceContainer devices1, devices2;  
devices1 = pointToPoint1.Install(nodes.Get(0), nodes.Get(1));  
devices2 = pointToPoint2.Install(nodes.Get(0), nodes.Get(2));
```

Different device be the connection between different flow
(node 0 is the client and node 1 & node 2 is the server)

```
Ipv4AddressHelper address1, address2;  
address1.SetBase("10.0.1.0", "255.255.255.0");  
address2.SetBase("10.0.2.0", "255.255.255.0");  
Ipv4InterfaceContainer interfaces1 = address1.Assign(devices1);  
Ipv4InterfaceContainer interfaces2 = address2.Assign(devices2);
```

In the SetBase , the number means that it would start allocating the IP address from "10.0.1.0"("10.0.2.0") using the subnet mask "255.255.255.0"

```
UdpEchoServerHelper echoServer1(99);  
ApplicationContainer serverApps1 = echoServer1.Install(nodes.Get(1));  
serverApps1.Start(Seconds(1.0));  
serverApps1.Stop(Seconds(10.0));  
  
UdpEchoClientHelper echoClient1(interfaces1.GetAddress(1), 99);  
echoClient1.SetAttribute("MaxPackets", UintegerValue(4));  
echoClient1.SetAttribute("Interval", TimeValue(Seconds(1.0)));  
echoClient1.SetAttribute("PacketSize", UintegerValue(1024));  
ApplicationContainer clientApps1 = echoClient1.Install(nodes.Get(0));  
clientApps1.Start(Seconds(2.0));  
clientApps1.Stop(Seconds(10.0));
```

(Taking the first server as example, using port 99)

In the *UdpEchoServerHelper*, the parameter represents the port number
And change the MaxPackets to 4(sending 4 udp packets)

(3) Execution

In the virtual box, after we can run the first.cc successfully, we cd into the /workspace/ns-3-allinone/ns-3-dev , and ./ns3 run scratch/112550018(the file), and we can get the result

```

cn2023-lab1@cn2023lab1-VirtualBox:~/workspace/ns-3-allinone/ns-3-dev$ ./ns3 run scratch/112550018.cc
At time +2s client sent 1024 bytes to 10.0.1.2 port 99
At time +2s client sent 1024 bytes to 10.0.2.2 port 98
At time +2.00481s server received 1024 bytes from 10.0.2.1 port 49154
At time +2.00481s server sent 1024 bytes to 10.0.2.1 port 49154
At time +2.00622s server received 1024 bytes from 10.0.1.1 port 49153
At time +2.00622s server sent 1024 bytes to 10.0.1.1 port 49153
At time +2.00962s client received 1024 bytes from 10.0.2.2 port 98
At time +2.01243s client received 1024 bytes from 10.0.1.2 port 99
At time +3s client sent 1024 bytes to 10.0.1.2 port 99
At time +3s client sent 1024 bytes to 10.0.2.2 port 98
At time +3.00481s server received 1024 bytes from 10.0.2.1 port 49154
At time +3.00481s server sent 1024 bytes to 10.0.2.1 port 49154
At time +3.00622s server received 1024 bytes from 10.0.1.1 port 49153
At time +3.00622s server sent 1024 bytes to 10.0.1.1 port 49153
At time +3.00962s client received 1024 bytes from 10.0.2.2 port 98
At time +3.01243s client received 1024 bytes from 10.0.1.2 port 99
At time +4s client sent 1024 bytes to 10.0.1.2 port 99
At time +4s client sent 1024 bytes to 10.0.2.2 port 98
At time +4.00481s server received 1024 bytes from 10.0.2.1 port 49154
At time +4.00481s server sent 1024 bytes to 10.0.2.1 port 49154
At time +4.00622s server received 1024 bytes from 10.0.1.1 port 49153
At time +4.00622s server sent 1024 bytes to 10.0.1.1 port 49153
At time +4.00962s client received 1024 bytes from 10.0.2.2 port 98
At time +4.01243s client received 1024 bytes from 10.0.1.2 port 99
At time +5s client sent 1024 bytes to 10.0.1.2 port 99
At time +5s client sent 1024 bytes to 10.0.2.2 port 98
At time +5.00481s server received 1024 bytes from 10.0.2.1 port 49154
At time +5.00481s server sent 1024 bytes to 10.0.2.1 port 49154
At time +5.00622s server received 1024 bytes from 10.0.1.1 port 49153
At time +5.00622s server sent 1024 bytes to 10.0.1.1 port 49153
At time +5.00962s client received 1024 bytes from 10.0.2.2 port 98
At time +5.01243s client received 1024 bytes from 10.0.1.2 port 99

```

Questions

1.Simulation&Emulation

The simulators are using mathematical models and algorithm to replicate the functionality of network devices, it provides a controlled environment for testing and analysis.

Comparing to the simulation, emulation are more alike running actual software on actual hardware, which provide much more realistic environments and results, also offer higher accuracy in representing real-world behaviors.

2.NS-3 problem

By default, ns-3 simulations use a fixed seed; if there is any randomness in the simulation, each run of the program will yield identical results unless the seed and/or run number is changed.

To obtain randomness across multiple simulation runs, you must either set the seed differently or set the run number differently. To set a seed, call *SeedManager::SetSeed (uint32_t)* at the beginning of the program; to set a run number with the same seed, call *SeedManager::SetRun (uint32_t)* at the beginning of the program.

Difficulties and Learned

In this section, I delve deeper into understanding networks through the utilization of NS-3. This involves exploring protocols, ports, devices, and interfaces using the network simulator. Additionally, I familiarize myself with the fundamental components when generating a flow.

The primary challenge lies in navigating the code, given the multitude of variables that require adjustment. Nevertheless, readability is maintained by referring to the documentation provided in NS-3 tutorials. The most significant hurdle is the considerable time invested in the setup process. Despite this, the complexity is somewhat alleviated compared to Lab1, making it a manageable task.